

Attosecond Pulses and Advanced Techniques Planned for LCLS-II

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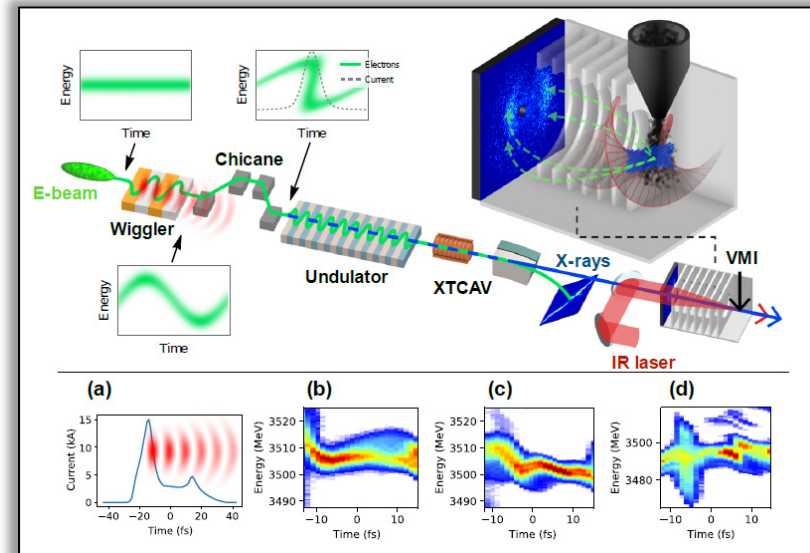
North American Particle Accelerator Conference

Lansing, Michigan

XLEAP Project at LCLS

- X-ray Laser-Enhanced Attosecond Pulse Generation

- Produce attosecond pulse at soft X-ray FEL
- Characterize the isolated attosecond pulses by angular streaking technique
- Experimental results and analysis have been reported by S. Li (TUYBA1).



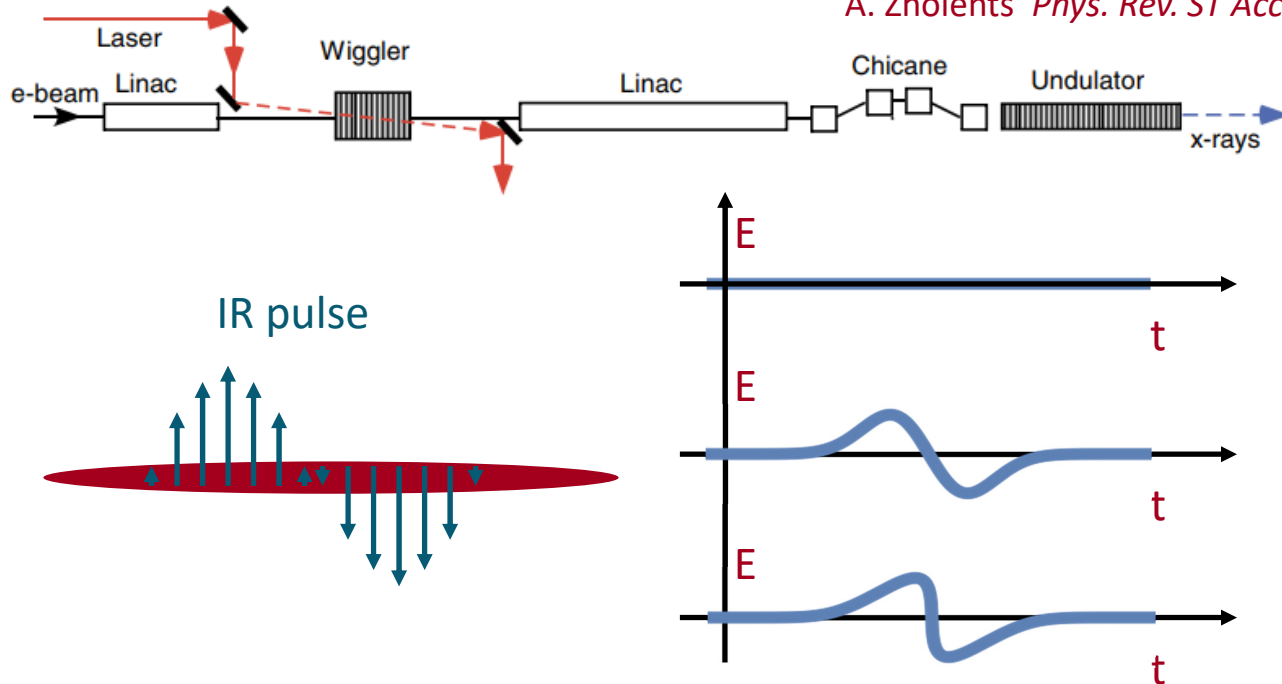
- XLEAP-II for LCLS-II:

- electron beam from copper linac
- more wiggler periods (from 6 to 30)
- variable-gap undulators for chirp-taper optimization

Basic Scheme in XLEAP

- Enhanced Self-Amplified Spontaneous Emission (E-SASE)

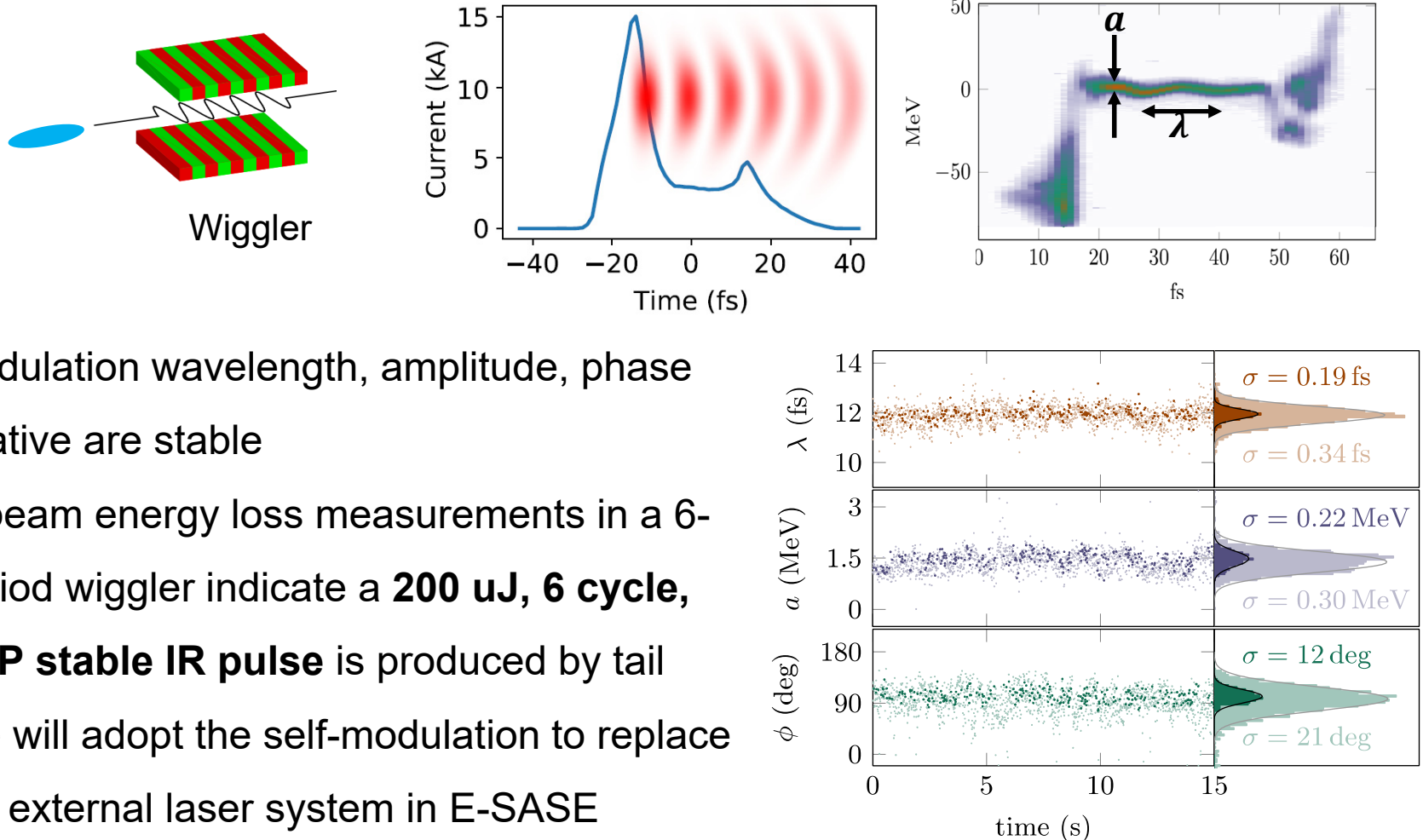
A. Zholents *Phys. Rev. ST Accel. Beams* 8, 040701 (2005)



- E-SASE can decrease the gain length by enhancing peak current, which makes it possible to produce sub-fs pulse at soft X-ray.

Basic Scheme in XLEAP (2)

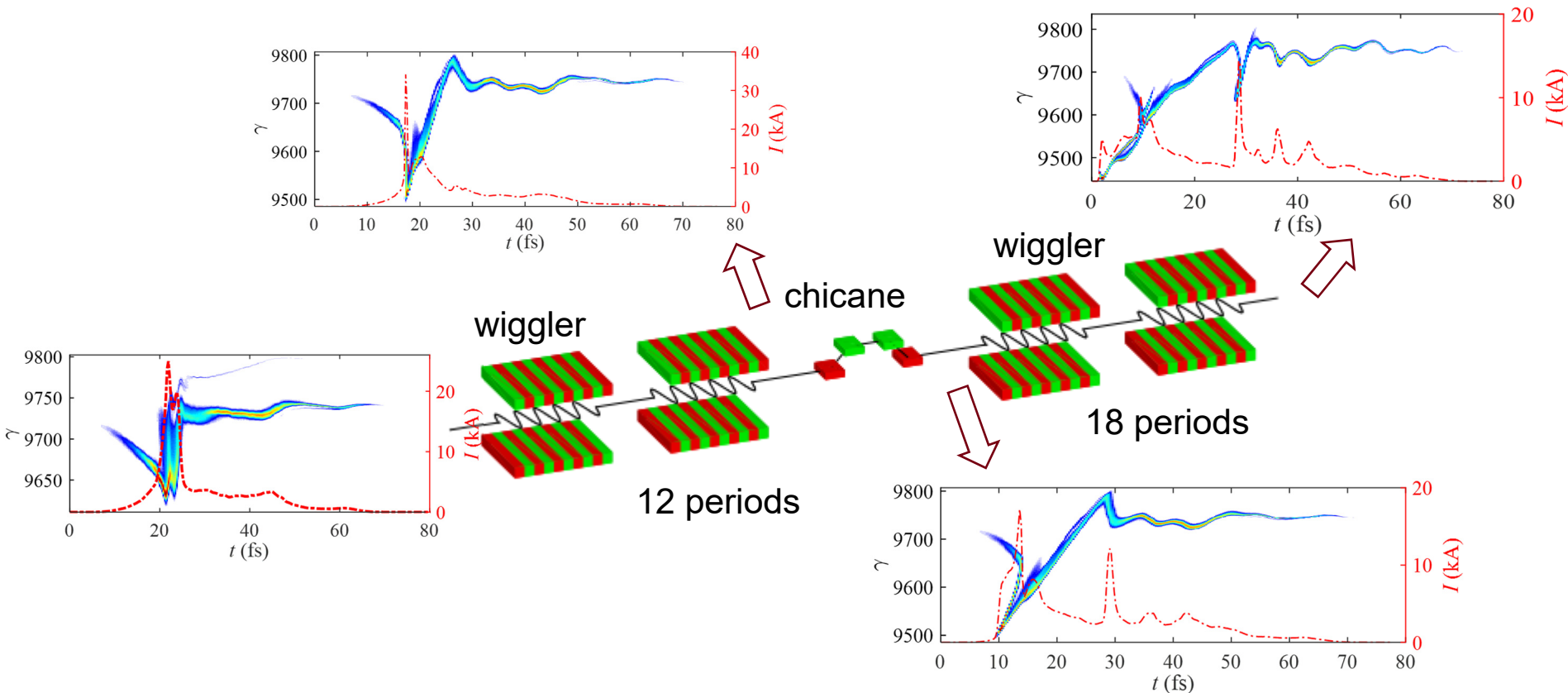
- Self-modulation in the wiggler



- Modulation wavelength, amplitude, phase relative are stable
- E-beam energy loss measurements in a 6-period wiggler indicate a **200 uJ, 6 cycle, CEP stable IR pulse** is produced by tail
- We will adopt the self-modulation to replace the external laser system in E-SASE

Two-Stage Self-Modulation in LCLS-II

- Due to the R56 of wiggler, the rotation of tail current horn limits the available periods for single-stage self-modulation
- We propose the two-stage scheme to increase the modulation amplitude



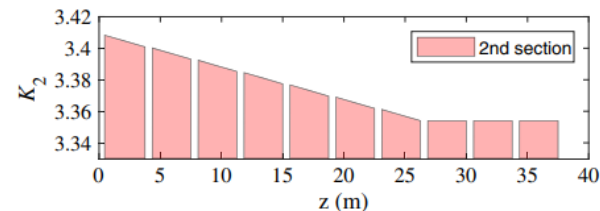
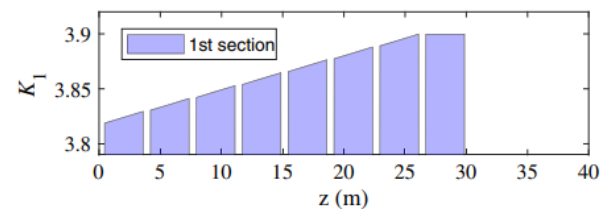
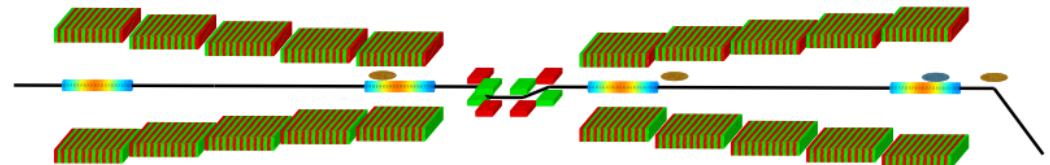
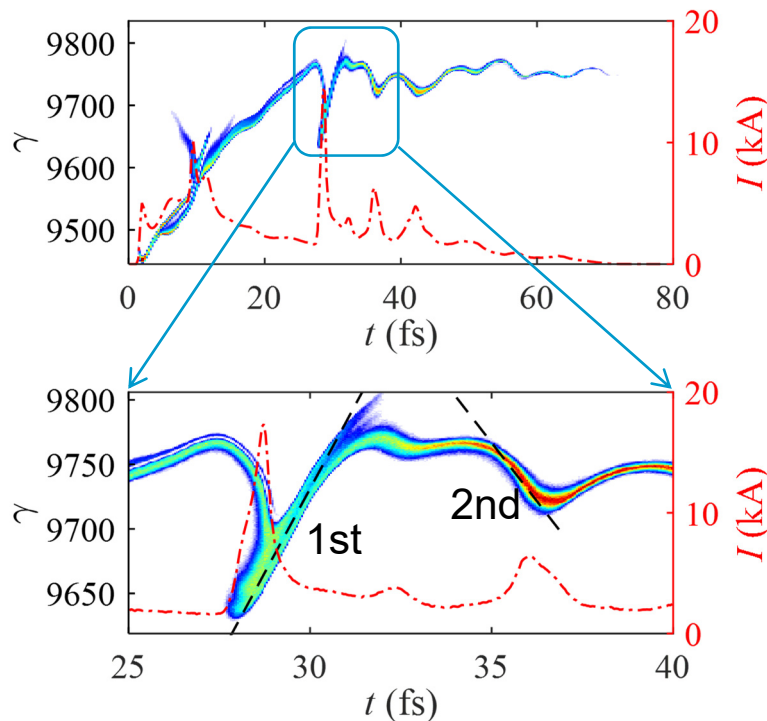
Two-Color Attosecond XFELs

- We use chirp-taper technique to produce two-color attosecond XFEL pulses

$$\frac{\lambda_u K(0)}{1 + K(0)^2/2} \frac{dK}{dz} = \frac{2\lambda_r}{c\gamma_r} \frac{d\gamma}{dt}$$

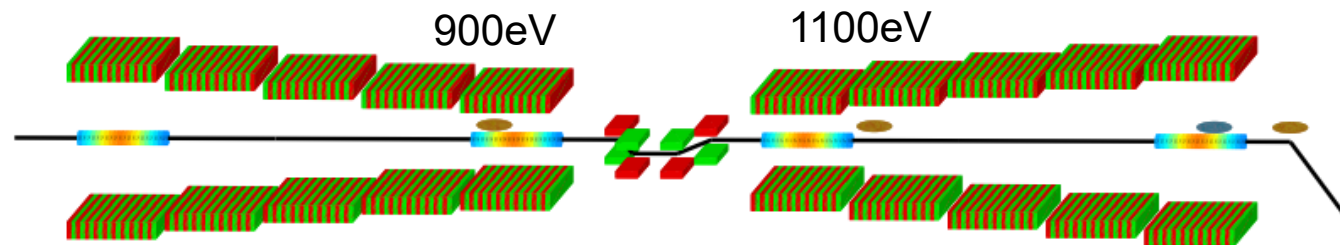
Chirp

Taper

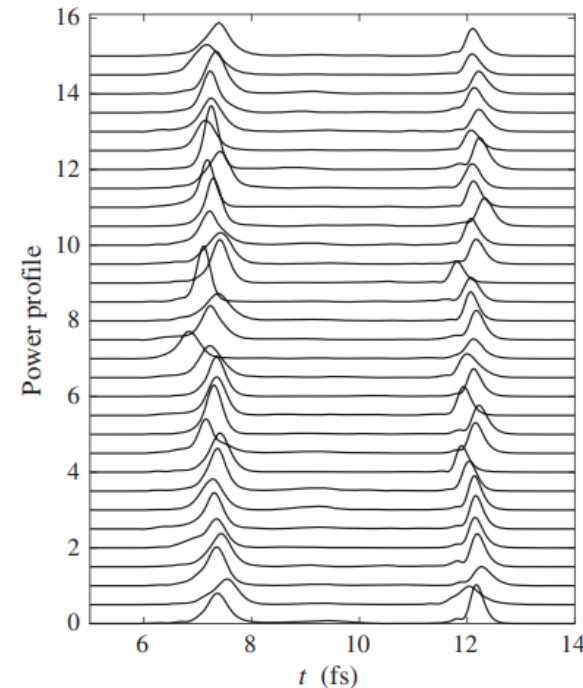
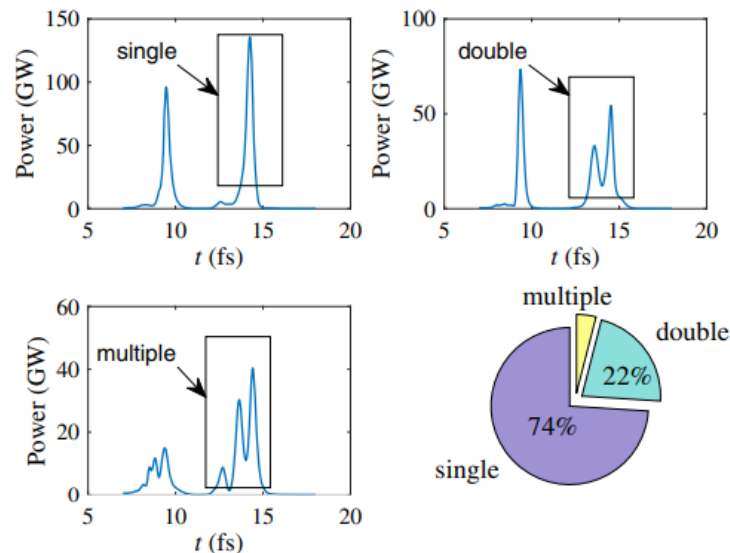


Two-Color Attosecond XFELs (2)

- Statistics of Two-color XFEL pulses



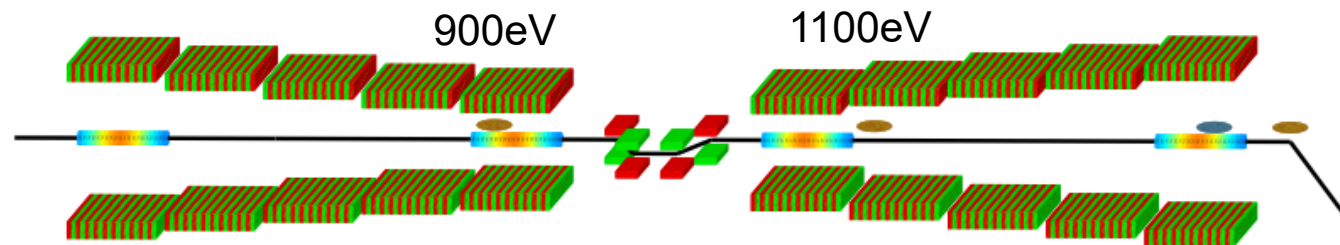
- With the optimized undulator taper, most of shots (~74%) have single-spike power profile in both colors.



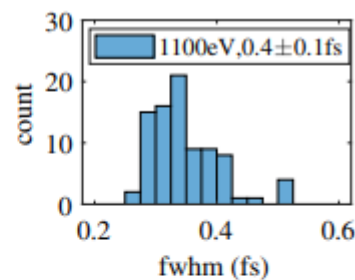
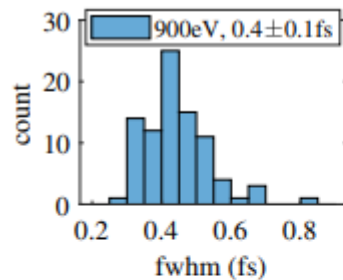
Typical
30 shots

Two-Color Attosecond XFELs (3)

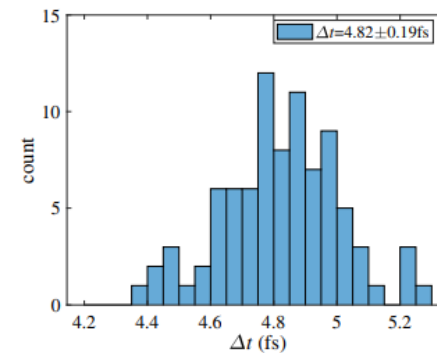
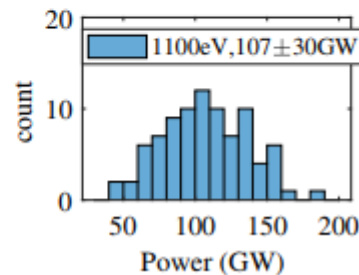
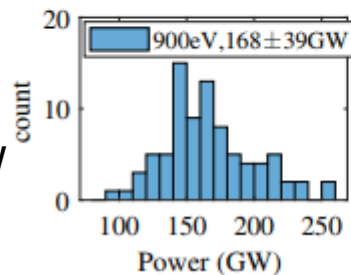
- Statistics of Two-color XFEL pulses



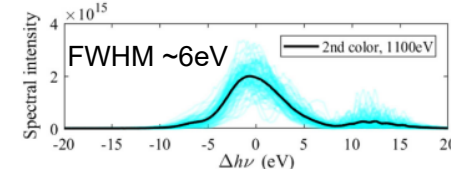
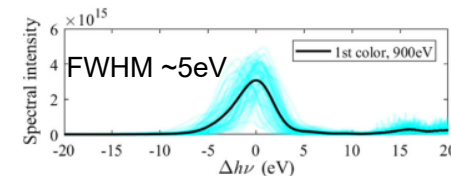
FWHM
~0.4fs



Power
>100GW



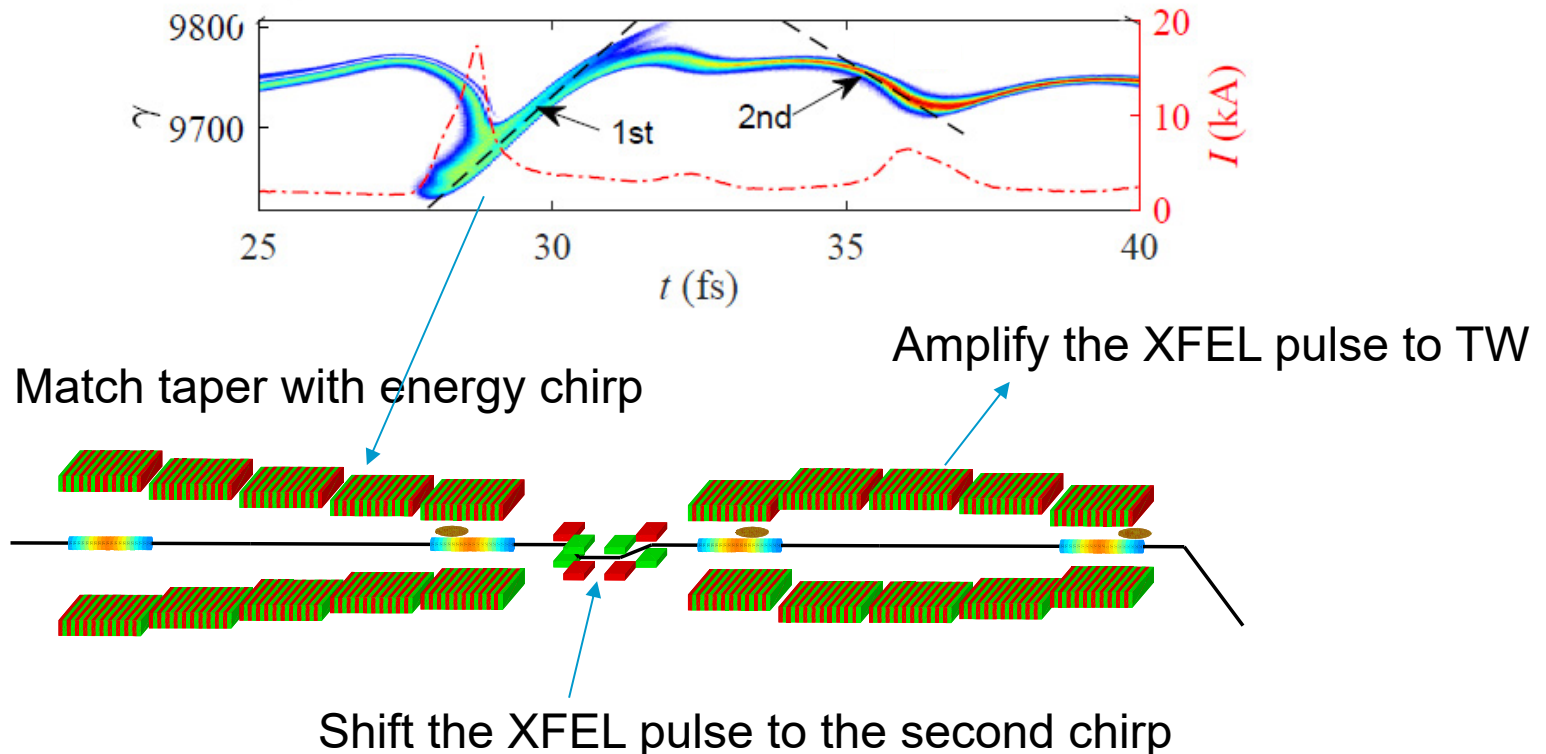
Time delay
Jitter 0.19fs



Spectra

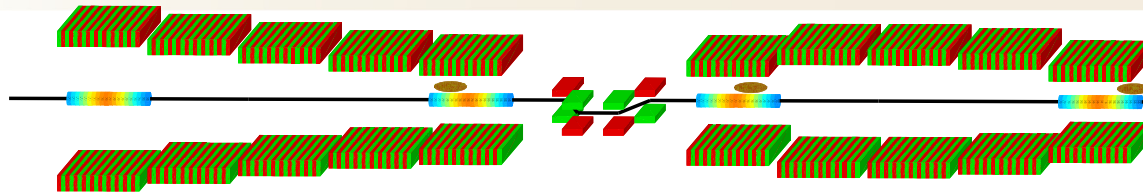
Terawatt Attosecond XFEL

- The two-stage undulators and chicane can be cascaded to achieve isolated attosecond TW XFEL pulse.



Terawatt Attosecond XFEL (2)

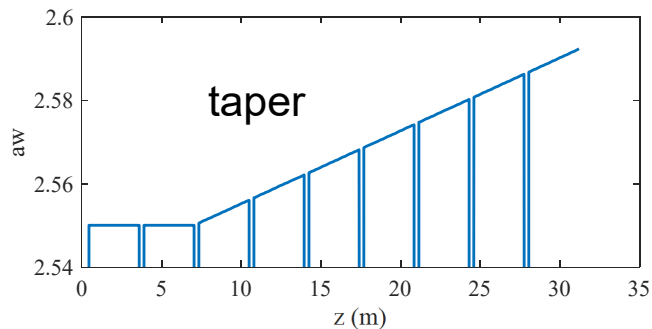
SLAC



Average power over 50 shots

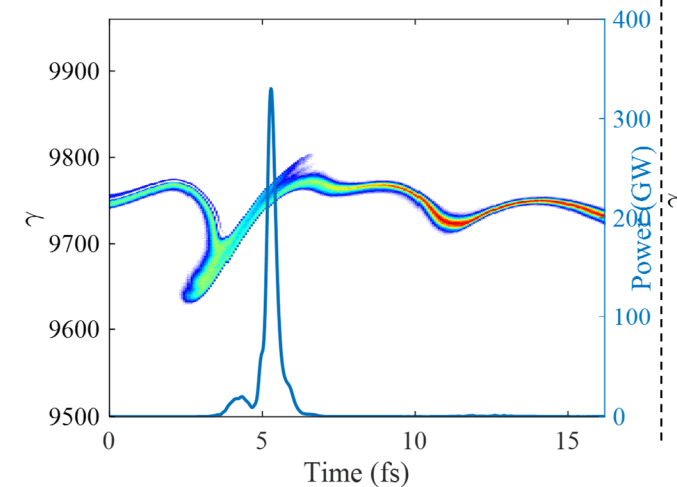
$$P = 1027 \pm 210 \text{ GW}$$

First stage:

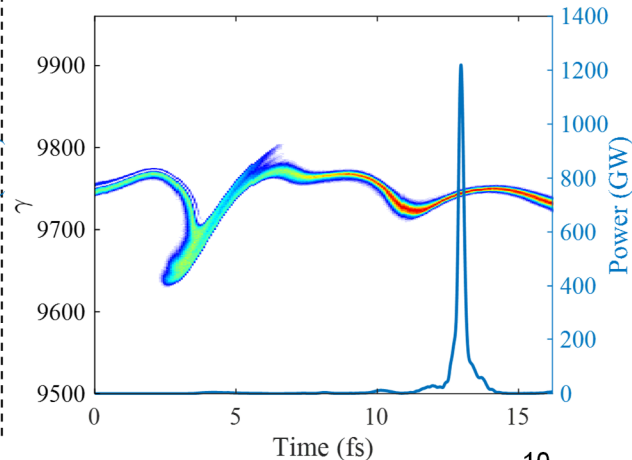
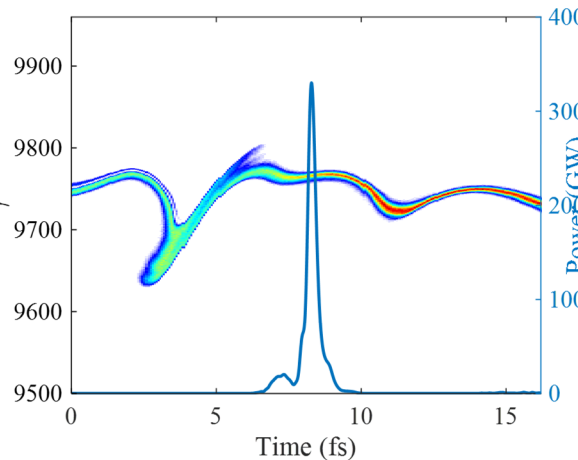
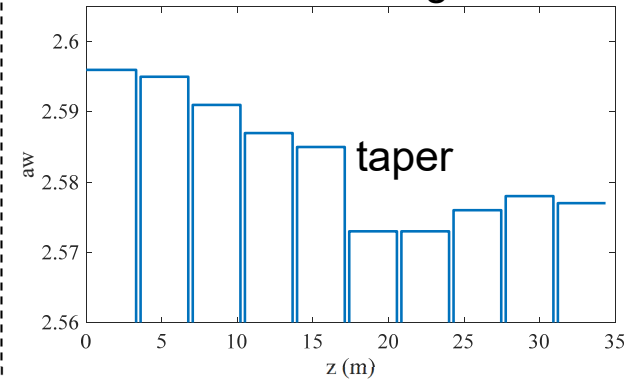


Chicane:

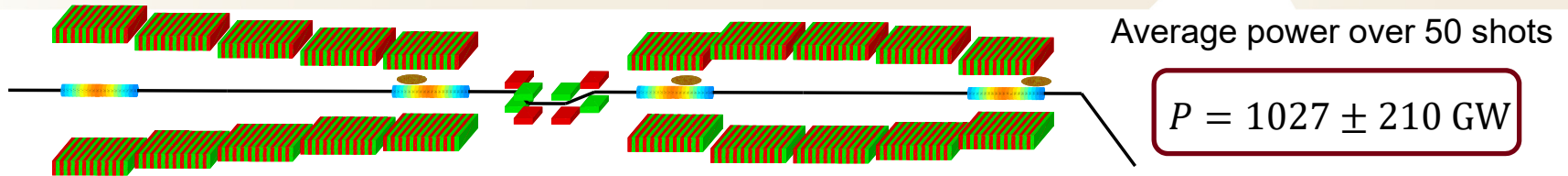
$$\Delta t = 3 \text{ fs}$$



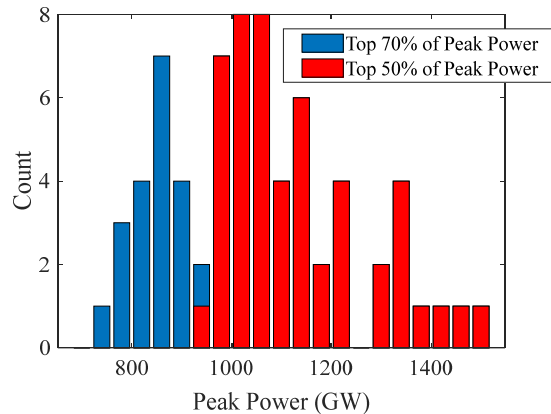
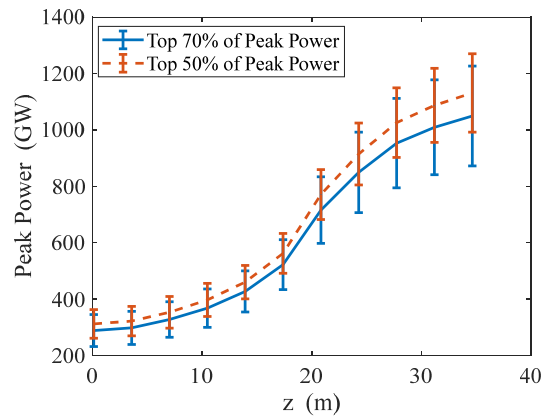
Second stage:



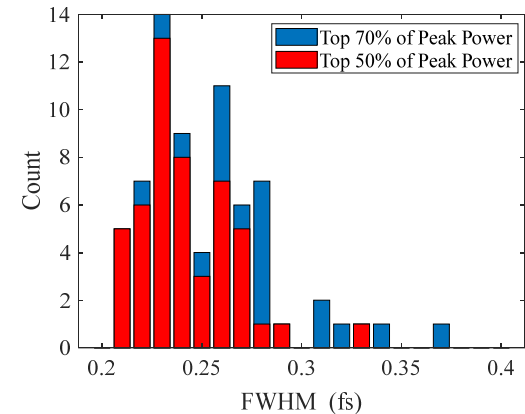
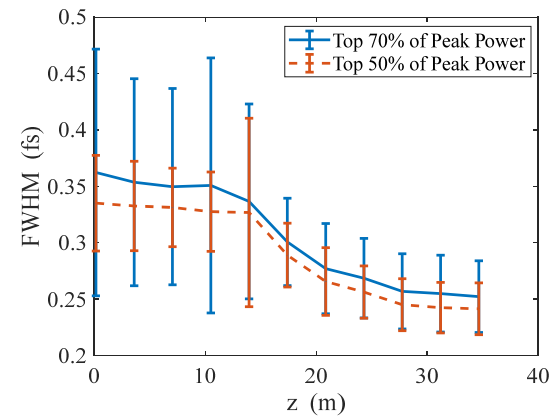
Terawatt Attosecond XFEL (3)



Peak power



Pulse duration



Summary

- E-SASE and self-modulation are applied to LCLS/LCLS-II to produce attosecond pulses at soft X-ray range.
- We propose the two-stage self-modulation scheme from the tail current horn of copper-linac beam in LCLS-II.
- Two-color attosecond XFEL can be generated by chirp-taper technique.
 - Single-spike power profile, Peak power $\sim 100\text{GW}$, FWHM duration $\sim 0.4\text{fs}$
 - Timing jitter $\sim 0.19\text{fs}$, Bandwidth $\sim 5/\sim 6\text{eV}$
- The two-stage undulator can be cascaded to amplify the XFEL pulse from the first stage through super-radiant to reach more than TW power.

XLEAP/XLEAP-II Projects



XLEAP Team:

SLAC

AD: A. Marinelli (project lead), J. Duris, J. MacArthur, Z. Zhang
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D. Bohler, M. Guetg, T. Maxwell, P. Baxevanis, Z. Huang
LCLS: M. Glowia, A. Fry, S. Vetter, P. Hering, D. Ray, T. Osipov,
R. Coffee, A. Miahnari, B. Smith, J. Hastings
PULSE: J. Cryan, P. Bucksbaum, E. Champenois, T. Driver

ANL

A. Zholents, J. Xu

LMU

M. Kling, P. Rosenberger

Funding:

- Panofsky Fellowship
- DOE-BES ADR program FWPs
10037 and 100497
- LCLS Operations

Principal Investigators



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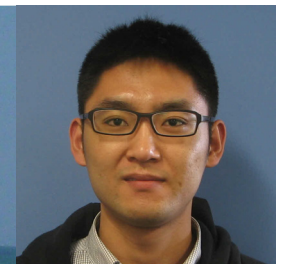


J. Cryan

Postdocs



J. Duris



Z. Zhang

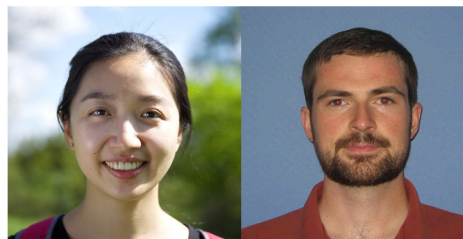


E. Champenois

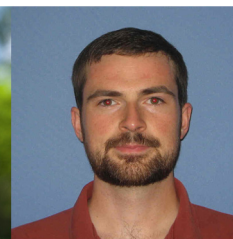


T. Driver

Grad Students



Siqi Li



J. MacArthur

Thanks for your attention!